Time Series Analysis

PARAMI UNIVERSITY

Time Series Analysis

Today's Topic: Moving Average Model

Dr. Si Thu Aung

in/drsithuaung/

https://sithuaung-ink.github.io/data-analyst-portfolio/

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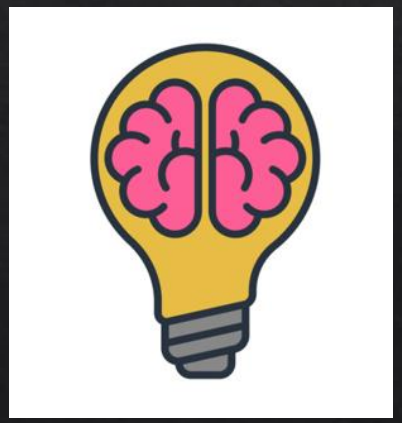
Content

Introduce Moving Averages



Practical Considerations

Moving Average Example

- Consider a window of size 3
- Compute at the center of the window
- Compute mean
- Move window and iterate
- 3-MA is a shorter time series

Date	y	3-MA
2026-01-1	23	NaN
2026-01-2	30	41.0
2026-01-3	70	43.3
2026-01-4	30	46.6
2026-01-5	40	31.6
2026-01-6	25	29.0
2026-01-7	22	22.6
2026-01-8	21	20.3
2026-01-9	18	33.0
2026-01-10	60	42.6
2026-01-11	50	NaN

Moving Average

- Moving average of order m (denoted m-MA):

$$Z_t = \frac{1}{m} \sum_{j=-k}^{j=k} y_{t+j}$$

- Where m=2k+1 is the size of the window, where k data points, either side of t, are included in the average.
- Each data point in the window receives equal weight, and the window is symmetric.

https://en.wikipedia.org/wiki/Moving_average

Moving Average

Even window size

- Often, the window size is selected to be the same as the seasonality to smooth out seasonal variation
- Example: Monthly data, yearly seasonality T=12
- With an even window, where do we compute the mean value?

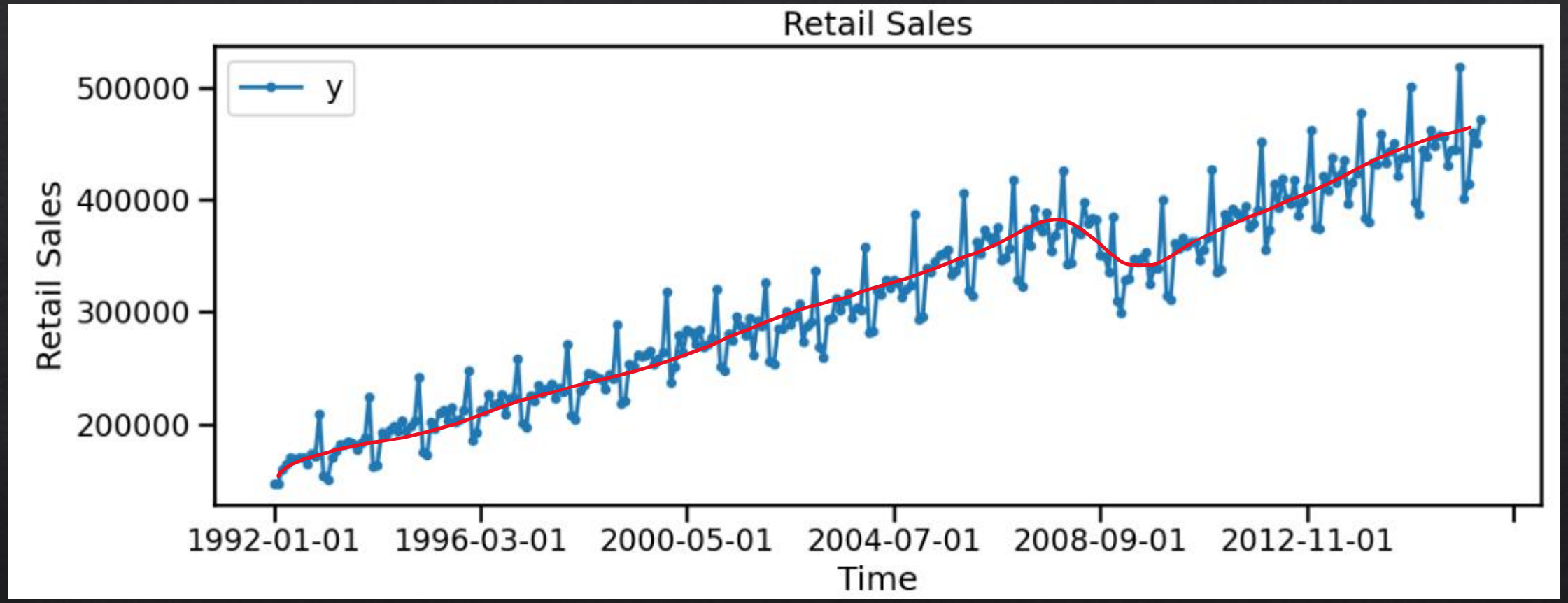
2025												2026	
Jan	Feb	March	April	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb

Even Window Size

- Compute 4-MA, which is 4 point moving average
- Compute 2×4-MA again
- 12-month seasonality data

Time Index	Middle Time Index	y	4-MA	2 × 4-MA
1		23		
	1.5			
2		70		
	2.5		38.7	
3		40		38.1
	3.5		37.5	
4		22		35.0
	4.5		32.5	
5		18		
	5.5			
6		50		

MA to Extract the Trend



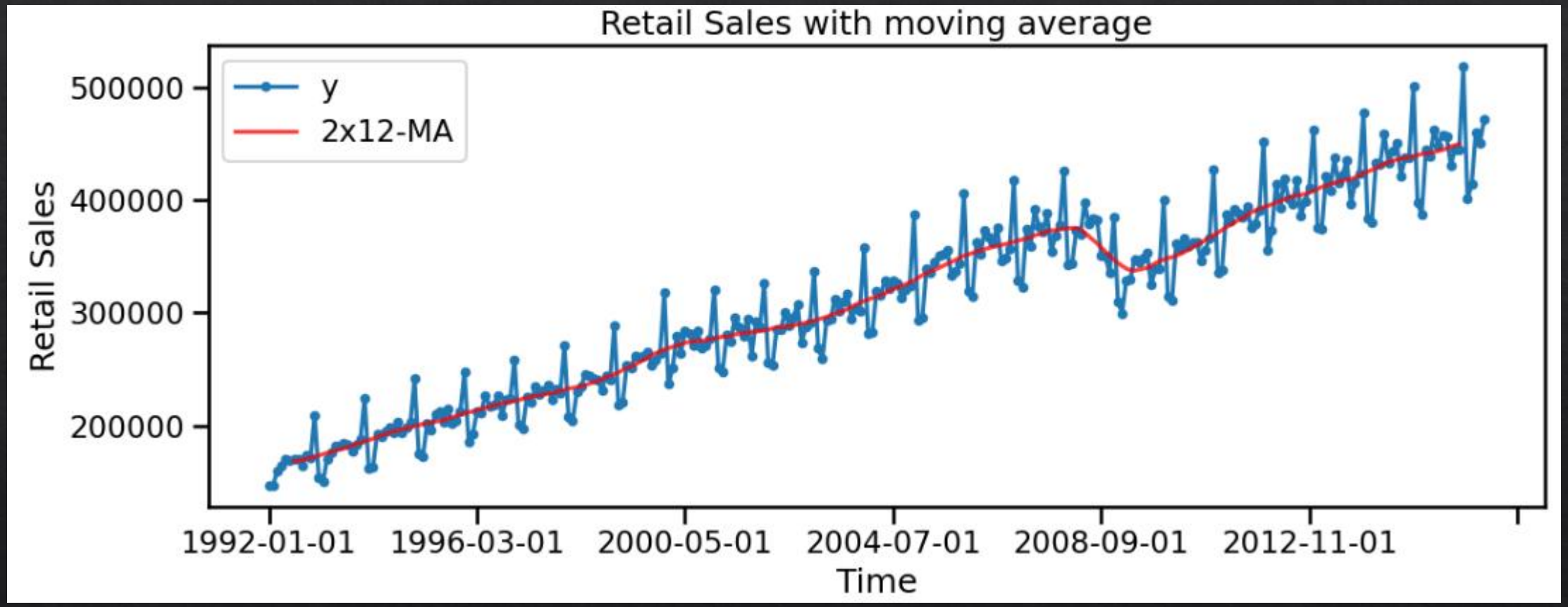
- How can we extract the trend?

MA to Extract the Trend

- What window size?
- Assume time series has seasonality with period T (e.g., T=12 for monthly data with a yearly seasonality)
- If T is odd: T-MA
- If T is even: use a 2 × T-MA
- This will smooth over the seasonality

Date	y	3-MA
2026-01-1	23	
2026-01-2	30	41.0
2026-01-3	70	43.3
2026-01-4	30	46.6
2026-01-5	40	31.6
2026-01-6	25	

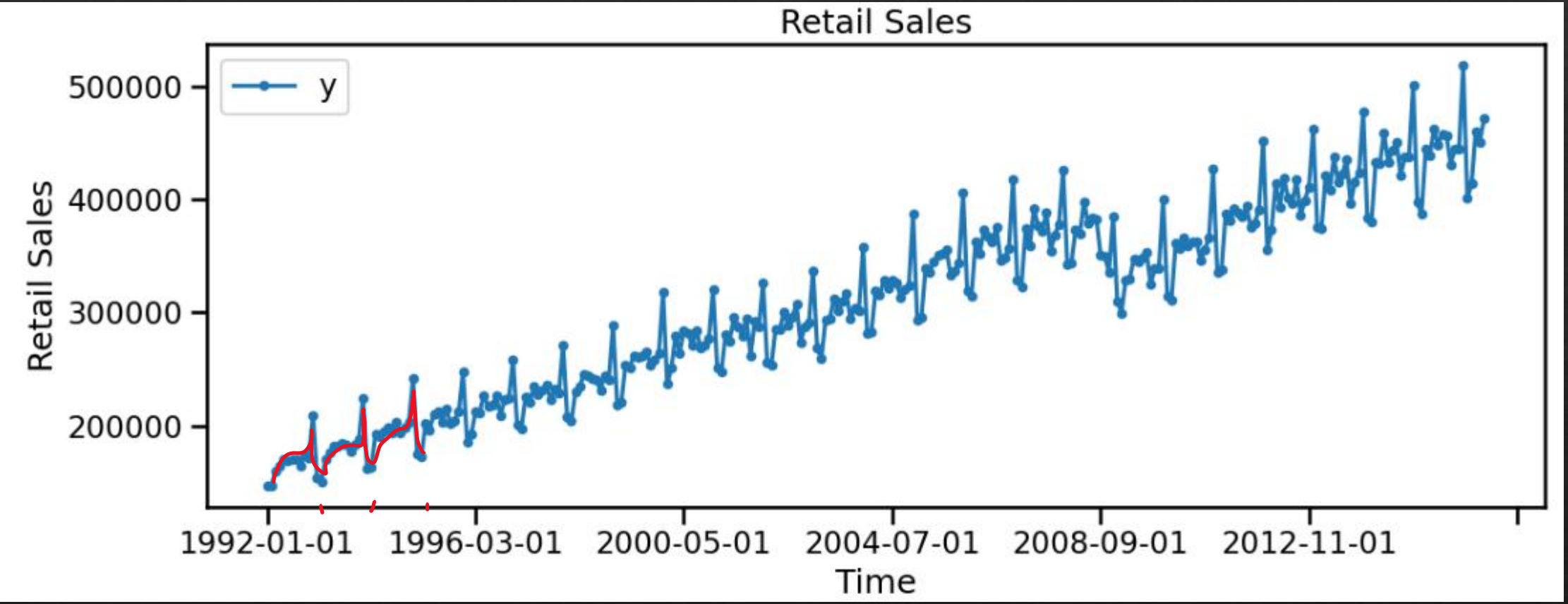
MA to Extract the Trend



MA to Extract the Trend

- What if no obvious seasonality?
- Visually inspect different windows sizes to ensure that the main trend is to ensure that the main trend is captured
- Too small a window: will capture noise and seasonality rather than the overall trend
- Too large a window: will over-smooth variations which might be included in the trend

MA to Extract the Seasonality



- How can we extract the seasonality?

MA to Extract the Seasonality

How can we extract the seasonality?

- Additive
- Multiplicative

Additive

$y(t) = trend(t) + seasonal(t) + residual(t)$

$Seasonal(t) = y(t) - trend(t) - residual(t)$

Multiplicative

$y(t) = trend(t) \times seasonal(t) \times residual(t)$

$Seasonal(t) = y(t) \times residual(t) / trend(t)$

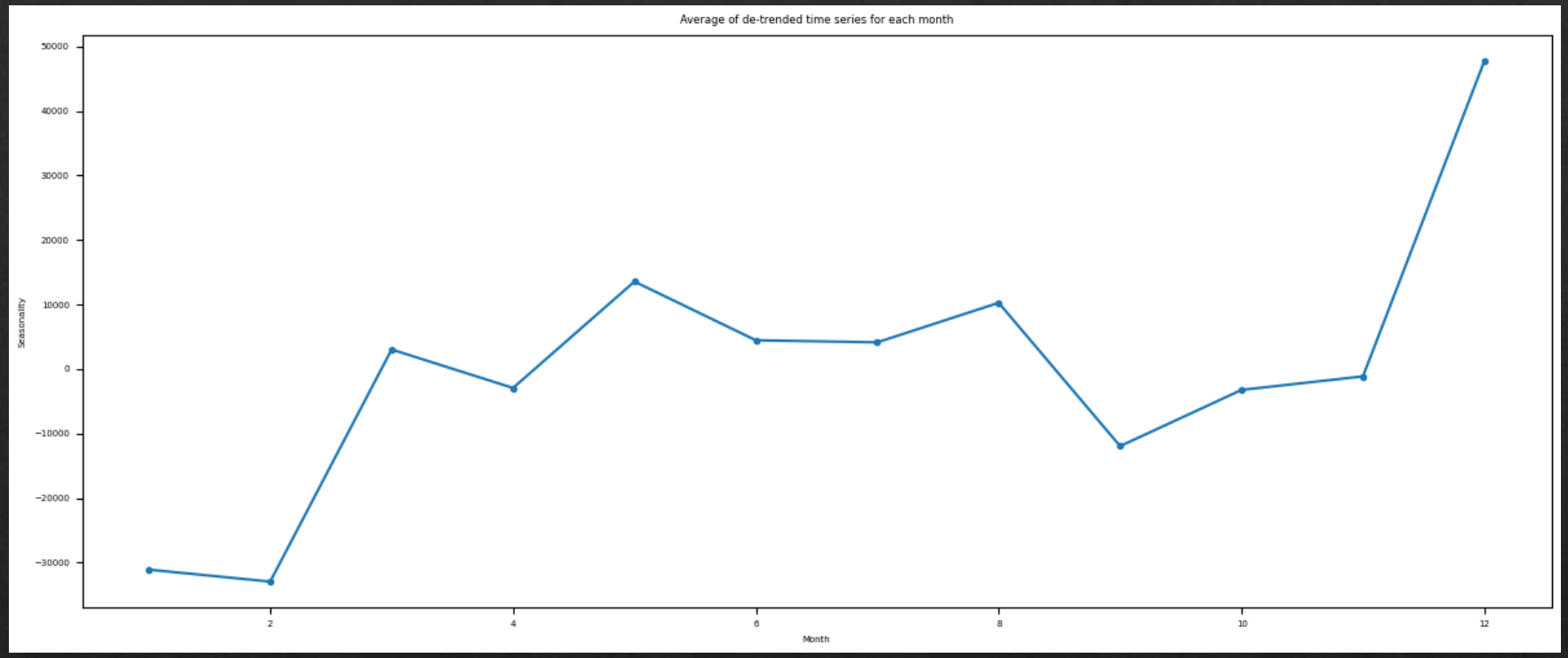
Trend: Estimate trend using moving averages

Residual: The data that remains after removing the model components (like trend and seasonality)

MA to Extract the Seasonality

- Identify the order of seasonality T -> Monthly or Yearly seasonality (if year T=12)
- Compute trend using T-MA (if odd) or 2×T-MA (if even)
- Detrend the data:
  1. if additive: y(t) – trend(t)
  2. if multiplicative: y(t)/ trend(t)
- Average the de-trended data over each seasonal index to remove noise (e.g., for monthly data, average all the months).

MA to Extract the Seasonality



- The seasonal pattern is fixed each year
- We can repeat the seasonal pattern each year to get seasonal(t)
- We can plot seasonal(t) alongside trend(t) and y(t)

Moving Average

pandas.DataFrame.rolling

```
DataFrame.rolling(window, min_periods=None, center=False, win_type=None,
on=None, closed=None, step=None, method='single')
```
[source]

Provide rolling window calculations.

Parameters:

window : *int, timedelta, str, offset, or BaseIndexer subclass*

Interval of the moving window.

If an integer, the delta between the start and end of each window. The number of points in the window depends on the `closed` argument.

If a timedelta, str, or offset, the time period of each window. Each window will be a variable sized based on the observations included in the time-period. This is only valid for datetimelike indexes. To learn more about the offsets & frequency strings, please see this link.

If a BaseIndexer subclass, the window boundaries based on the defined `get_window_bounds` method. Additional rolling keyword arguments, namely `min_periods`, `center`, `closed` and `step` will be passed to `get_window_bounds`.

min_periods : *int, default None*

Minimum number of observations in window required to have a value; otherwise, result is `np.nan`.

For a window that is specified by an offset, `min_periods` will default to 1.

For a window that is specified by an integer, `min_periods` will default to the size of the window.

Moving Average

```
statsmodels.tsa.seasonal.seasonal_decompose(
    x,
    model='additive',
    filt=None,
    period=None,
    two_sided=True,
    extrapolate_trend=0
)                                                    [source]
```

Seasonal decomposition using moving averages.

Parameters

`x : array_like`

Time series. If 2d, individual series are in columns. x must contain 2 complete cycles.

`model : {"additive", "multiplicative"}, optional`

Type of seasonal component. Abbreviations are accepted.

`filt : array_like, optional`

The filter coefficients for filtering out the seasonal component. The concrete moving average method used in filtering is determined by two_sided.

Moving Average

Summary

- Moving averages can be used to extract the trend.
- Seasonality can be extracted by detrending and averaging over a known seasonal index.